

YUVRAJ JAISWAL

AI / LLM Engineer · Agentic Systems · RAG Pipelines · LangChain / LangGraph · Full Stack

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PROFESSIONAL SUMMARY

AI / LLM Engineer with almost 3 years of production experience building LangChain / LangGraph-orchestrated agentic systems, RAG pipelines, and cloud-native Python backends on AWS. Shipped HyphenView — an AI Text-to-SQL platform with 65% enterprise adoption among 50+ users — and engineered a LangGraph multi-agent autonomous job-hunting system with real-time tool-calling and embedding-based skill matching. Experienced across the full LLM lifecycle: prompt engineering, vector store design, embedding strategy, multi-step agent orchestration, observability, and production deployment. Strong full-stack capability (React.js, FastAPI, PostgreSQL) enabling end-to-end ownership from model integration to user-facing UI.

TECHNICAL SKILLS

AI / LLM & Agents: LangChain, LangGraph, OpenAI GPT-4o, Anthropic Claude, Agentic Workflows, ReAct Loop, Tool-Calling, Multi-step Reasoning, Prompt Engineering, Prompt Chaining, Context Injection, Output Parsing

RAG & Vector Search: FAISS, ChromaDB, Pinecone (exploring), RAG Pipelines, Hybrid Retrieval (semantic + keyword), Embedding Models (text-embedding-ada-002), Chunking & Indexing Strategy, RAGAS Evaluation

Backend & APIs: FastAPI, Flask, REST APIs, Asyncio, JWT, RBAC, Redis Caching, Pydantic, Structured Logging, Modular Architecture

Cloud & DevOps: AWS (S3, Lambda, DynamoDB, ECS, CloudWatch, Bedrock), Docker, Docker Compose, CI/CD, Nginx

Databases: PostgreSQL, MongoDB, MySQL, Vector Stores (FAISS, ChromaDB)

Frontend: React.js, Next.js, Redux, Tailwind CSS, REST integration

Languages: Python, TypeScript, JavaScript, SQL

Practices: LLM Evaluation, Code Reviews, Pytest, API Documentation, Modular Design, Technical Documentation

PROFESSIONAL EXPERIENCE

AI / LLM Engineer · **Erasmith Pvt Ltd**

Feb 2024 – Present

Agentic AI & LLM Systems

- Designed and shipped LangGraph and LangChain-orchestrated multi-step agents over structured and unstructured enterprise data, including a production natural-language-to-SQL pipeline serving 50+ enterprise users with 65% adoption within the first quarter post-launch.
- Built ReAct-loop agentic workflows with tool-calling capabilities — enabling agents to query databases, call external APIs, and self-correct on intermediate outputs; integrated AWS S3 for document storage and Lambda for serverless trigger workflows.
- Integrated GPT-4o and Claude APIs into production backends with prompt engineering layers: schema-aware context injection, structured output parsing, and fallback handling for high-reliability inference.
- Implemented LLM observability via structured logging and AWS CloudWatch dashboards, tracking token usage, response latency, error rates, and inference cost across all production services — enabling data-driven prompt and infra tuning.
- Deployed GenAI services on AWS ECS (containerised via Docker) with automated CI/CD pipelines, maintaining 99.9% uptime across 100+ production application instances.

RAG Pipeline Architecture

- Architected RAG pipelines using FAISS and ChromaDB as vector stores with text-embedding-ada-002; documents persisted to AWS S3 and indexed on-demand via Lambda triggers for cost-efficient, serverless ingestion.
- Implemented chunking, indexing, and hybrid retrieval strategies (semantic + keyword) for long-form enterprise corpora, reducing hallucination rate on domain-specific query sets.

- Tuned chunk size, overlap, and retrieval parameters to optimize precision and GPT-4 context window utilisation, measurably improving answer grounding on production queries.

Backend, Cloud & Engineering

- Developed 80+ async FastAPI endpoints serving AI workflows with JWT authentication, RBAC, Redis caching, and Pydantic validation — sustaining sub-200ms p95 latency under concurrent load.
- Optimised PostgreSQL queries and indexing, cutting dashboard load time from 12s to 6.6s (45% improvement); persisted session state and job metadata in DynamoDB for stateless Lambda-backed workflows.
- Built and maintained React.js frontends integrated with LLM-backed APIs, enabling real-time query feedback, result visualisation, and responsive UX across AI-powered products.
- Led code reviews, maintained API and architecture documentation, and enforced Python best practices (type hints, structured logging, modular design) across team contributions.

KEY PROJECTS

HyphenView — AI Text-to-SQL Platform — *Python · FastAPI · LangChain · GPT-4o · PostgreSQL · ChromaDB · AWS S3 · Docker · React.js*

- Built an AI-native analytics platform enabling business users to query databases in natural language — 65% adoption among 50+ enterprise users within Q1 post-launch.
- Designed a multi-stage LangChain pipeline: schema-aware prompt construction → GPT-4o SQL generation → validation → result formatting, with error-correction loops for malformed queries.
- Used ChromaDB to store schema embeddings and example query pairs for few-shot retrieval; document assets stored in AWS S3 with signed URL access for security.
- Shipped React.js frontend with real-time query feedback and result visualisation; deployed via Docker behind Nginx on AWS ECS with CloudWatch logging.

GitHub : <https://github.com/YuvrajJais9257>

Smart Job Hunter — Autonomous LLM Multi-Agent System — *Python · FastAPI · LangGraph · LangChain · OpenAI · Redis · ChromaDB · Docker*

- Architected a LangGraph-powered autonomous agent using a ReAct loop with tool-calling to search live job listings, extract structured data, score role fit, and rank results by relevance.
- Built custom LangChain tools for JD parsing and embedding-based skill matching against a ChromaDB vector store; implemented async Redis job queue for multi-step workflows without API-layer blocking.
- Wrote Pytest integration tests covering all tool-calling paths and state transitions; documented agent architecture and full tool registry with API specs.

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AI Shopping Assistant — Tool-Using Generative Agent — *Python · FastAPI · LangChain · OpenAI · PostgreSQL · Docker*

- Built a tool-using generative agent that accepts budget and shopping intent, calls product search APIs, compares options, and produces ranked purchase recommendations with reasoning traces.
- Designed tool registry using LangChain's tool-calling interface; persisted product catalogs and session state in PostgreSQL with connection pooling; deployed via Docker Compose.

GitHub : <https://github.com/YuvrajJais9257>

EDUCATION

B.Tech in Computer Science — AI Specialization

2020 – 2025

Galgotias College of Engineering and Technology, Noida

CERTIFICATIONS

- AI / LangChain & Agentic AI — Udemy (RAG, LangGraph, Prompt Engineering, Agentic Workflows)
- Docker Mastery — Bret Fisher (Udemy)
- 100 Days of Code — Python — Angela Yu (Flask, PostgreSQL, Selenium, Data Science)
- Data Structures & Algorithms — Coding Ninjas
- Full Stack Web Development — Hitesh Choudhary

